CHEN-YE HONG

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Research Interests

Brain–computer interfaces for adaptive assistance, with an interest in how users and systems adapt to each other through neural and physiological feedback, and how users can guide the assistance they receive.

Education

National Taiwan University (NTU) Sep. 2022 – Jan. 2026
B.S. in Mechanical Engineering Taipei, Taiwan
GPA: 4.16/4.30 | Graduation Rank: 3/146

National Tsing Hua University (NTHU) Sep. 2021 – Jun. 2022 (Transferred to NTU)
Undergraduate Studies in Power Mechanical Engineering Hsinchu, Taiwan
GPA: 4.16/4.30 | Department Rank: 1/116

Publications

†Equal contribution (co-first authors).

- **C.-Y. Hong**†, C.-H. Chou†, Y.-J. Wang, K.-L. Wu, S.-J. Yang, C.-H. Chang, and D.-R. Li (2026). "An experimental cadaveric limb motion platform for detecting altered mechanical responses under small-joint defect conditions." Medical Engineering & Physics, 147(9), 095006. [DOI]
- C.-Y. Hsieh†, **C.-Y. Hong**†, J.-Z. Li†, P.-C. Yang†, Y.-C. Hsu, C.-H. Tsai, C.-H. Hsu, and Y.-T. Wang (2026). "From Movement to Manipulation: Multimodal Control of a Mobile Assistive Robot." Proceedings of the 14th International Winter Conference on Brain–Computer Interface (BCI 2026), IEEE. (Oral presentation) [DOI]

Submitted Manuscripts

- **C.-Y. Hong**†, I. C. Lee†, C.-C. Deng, J. P. Hsu, and Y.-T. Wang. "Competitive Play, Verified by the Body: A Multiplayer EEG/EMG VR Motor Game." Submitted to the 15th International Winter Conference on Brain–Computer Interface (BCI 2027).

Research Experience

RehabVerse: VR Motor Rehabilitation with Adaptive Scoring Jun. 2026 – Sep. 2026
Research Assistant (Full-time), BCI Lab, Academia Sinica | Dr. Yu-Te Wang

- Analyzed mu-band event-related desynchronization (ERD) during executed and imagined movements in 6 healthy participants, observing contralateral ERD during executed movements.
- Developed stage-based analyses of EEG lateralization to assess responses following score boosts and led manuscript writing and figure preparation.

VR Motor Rehabilitation with Human and AI Competition Jun. 2026 – Present
Research Assistant (Full-time), BCI Lab, Academia Sinica | Dr. Yu-Te Wang

- Conducted EEG experiments using the RehabVerse platform in a two-session study involving 22 healthy participants to compare solo, human-opponent, and AI-opponent conditions during executed and imagined movements.

Multichannel EEG Headset Development Jun. 2026 – Present
Research Assistant (Full-time), BCI Lab, Academia Sinica | Dr. Yu-Te Wang

- Designed and fabricated a six-channel EEG headset for a planned ALS communication interface combining SSVEP-based selection and eye tracking.
- Designing an EEG headset for use with AR glasses to record central and occipital activity, with a rotatable central electrode mount and interchangeable carriers for different head shapes.

LLM-Driven Robot Pet with State-Dependent Interaction Aug. 2026 – Present

Independent Research | GitHub

- Developed a MuJoCo robot pet that combines VLM-derived human behavioral cues with simulated internal states to guide LLM-based behavior selection.
- Built an LLM-driven loop that iteratively refines reward functions using training and evaluation feedback to teach a simulated robot to step over door thresholds.

Multimodal Control for an Assistive Mobile Manipulator Jul. – Aug. 2025

Summer Research Intern, BCI Lab, Academia Sinica | Dr. Yu-Te Wang

- Designed and 3D-printed the mobile base of an EEG-controlled assistive robot and supported offline and online EEG experiments.
- Implemented MediaPipe-based control of an open-source 6-DOF arm, using gestures to select joints and lateral hand motion to control rotation for a system that completed 2 object-retrieval tasks.

Human-Centric Automated Driving: VR-Based Passenger Emotion Assessment Sep. – Dec. 2025

Student Researcher (Part-time), BCI Lab, Academia Sinica | Dr. Yu-Te Wang

- Developed a Unity-based VR driving testbed to examine passenger valence and arousal across driving styles and road environments using EEG and self-reports.

Biomechanical Testing Platform for Small-Joint Evaluation Jan. 2024 – Oct. 2025

Undergraduate Researcher, SAM Lab, NTU | Prof. Dian-Ru Li

- Designed a cadaveric limb motion platform for synchronized force measurement and optical tracking to assess mechanical changes under controlled joint defects.
- Used PCA and k-means clustering to group marker trajectories and quantified regional deformation through apparent in-plane strain analysis.

Interactive VR Simulator for Cardiac Catheterization Training Jul. 2023 – Aug. 2024

Undergraduate Researcher, Advanced Medical Device Lab, NTU | Prof. Hao-Ming Hsiao

- Developed a Unity-based VR prototype for interactive guidewire positioning and balloon-driven stent expansion.

Industry Experience

Mechanical Engineering Intern, HP Inc. Jul. 2024 – Jun. 2025

- Redesigned a laptop hinge to improve thermal performance and developed a modular SSD shielding cap adopted across 3 product lines, cutting replacement time by 86% compared with similar company designs.

Leadership Experience

Founder & President Sep. 2024 – Jan. 2026

NTU Mechanical Engineering International Compass

- Founded a student organization to expand access to overseas study information and led a team of up to 22 members that conducted 30+ alumni interviews and organized forums reaching 300+ attendees.

Honors and Awards

Dean's List Award, NTU (top 5% of class) Fall 2024, Spring 2025

Academic Achievement Award, NTHU (top 5% of class) Fall 2021, Spring 2022

Technical Skills

Programming	Python, C# (Unity), MATLAB
EEG Analysis	MNE-Python, EEGLAB
ML & Vision	PyTorch, OpenCV, MediaPipe, YOLOv8
Robotics & AI	MuJoCo, RL (PPO), LLMs, VLMs
Hardware & VR	SolidWorks, Autodesk Inventor, 3D printing (FDM/SLA), Unity